

Cambricon

AI Infrastructure Architecture Evolution

2015-2026

Silicon generations · Memory and interconnects
Conference disclosures · Systems and software

A generation-by-generation synthesis of conference disclosures and primary technical sources.
Evidence cutoff: 28 September 2026.
Complete English edition, paired with a full Simplified Chinese report.

Contents

01 Scope and architectural synthesis	3
02 Research lineage: what the architecture conferences establish	3
03 Commercial generations: MLU100, 270 and 290	4
04 MLU370: chiplets, fusion and board differentiation	4
05 Cloud accelerator and board tables	5
06 MLU-Link, systems and the networking roadmap	9
07 Terminal IP, edge products and software continuity	13
08 Normalized metrics and deployment implications	17
09 Conference index and product mapping	18
Source map	21
References	22

01 Scope and architectural synthesis

The report covers Cambricon's publicly documented AI infrastructure from 2015 to 28 September 2026, with DianNao-era research as a pre-company baseline. Deep coverage follows Siyuan/MLU cloud accelerators, memory, MLU-Link and NeuWare. Terminal IP and the MLU220 edge branch explain technology continuity. No proprietary server CPU, merchant DPU or shipping AI NIC is inferred merely from the company's broader AI ambitions.

The strongest continuity is programmable domain specialization: a machine-learning instruction interface, tensor execution, explicit data movement and a software platform spanning cloud and edge. Commercial evolution adds larger SoCs, training support, chiplets and multi-device communication. The public academic record is much richer than the disclosure of recent commercial implementations. Those two evidence streams should inform each other without being treated as an exact RTL lineage.

02 Research lineage: what the architecture conferences establish

DianNao at ASPLOS 2014 and DaDianNao at MICRO 2014 established early neural-acceleration research. PuDianNao at ASPLOS 2015 broadens the machine-learning workload scope; ShiDianNao at ISCA 2015 moves vision processing toward the sensor. The author's institutional page explicitly reserves the DianNao names for those academic designs. They are therefore recorded as research predecessors, not early commercial Siyuan SKUs.

The ISCA 2016 Cambricon ISA paper addresses the gap between fixed-function neural engines and general-purpose instruction streams. A domain-specific ISA can express common neural operations while preserving programmability. That is a useful lens for understanding later MLU software, but the published prototype's process, area and performance cannot be transplanted into a commercial MLU table without a documented mapping.

Cambricon-X at MICRO 2016 and Cambricon-S at MICRO 2018 address sparse and irregular networks. Cambricon-F at ISCA 2019 studies a hierarchical, self-similar programming architecture; Cambricon-Q at ISCA 2021 targets efficient training. The research progression spans sparsity, programmability and training precision. It establishes engineering priorities, not proof that an identically named mechanism ships in every generation.

Cambricon-F recursively decomposes operations through nodes sharing an instruction model and a hierarchical memory organization. The proposal aims to reduce the programming burden across scales. Architectural interpretation: a uniform interface does not eliminate placement, reduction or congestion costs; it changes where they are handled. The paper discusses those challenges explicitly. Its research instances F1/F100 are not MLU100 or MLU270 product names.

The 2024 Cambricon-LLM research proposes an NPU plus specially designed flash with on-die processing for low-batch edge inference. It is a useful example of capacity/bandwidth co-design, but is not a disclosed production implementation of a Siyuan server accelerator. Flash capacity, DRAM KV storage and compute-side bandwidth are distinct resources.

03 Commercial generations: MLU100, 270 and 290

Siyuan100 introduced the cloud accelerator line in 2018. The 270 generation adds MLUv02, a NoC connecting up to sixteen tensor cores, hardware data compression and video/image functions. Its official page explicitly identifies 128 INT8 TOPS for nonsparse models, alongside 256 INT4 TOPS and 64 INT16 TOPS. Precision changes must not be described as architectural speedups on the same workload.

Siyuan290 is the training-oriented branch, publicly exhibited with MLU290-M5 modules and the XuanSi1000/MLU-X1000 system at WAIC 2021. Its importance is the transition from a predominantly inference product story to multi-accelerator training and system software. This edition does not fill missing board-matched HBM, power or floating-point rates from secondary product lists. A training product's existence and an exact implementation specification are separate claims.

04 MLU370: chiplets, fusion and board differentiation

Siyuan370 combines two AI compute chiplets in a 7 nm product with 39 billion transistors. MLUarch03 adds tensor-unit improvements, the Supercharger convolution mechanism and hardware support for multi-operator fusion. LPDDR5 and MLU-Link expand the memory and multi-device design space. Chiplet integration is a packaging boundary within a chip; a dual-chip board is a separate boundary and must not be counted as the same thing.

The current official specifications are particularly informative: X4 and X8 both list 96 TFLOP/s BF16 and 256 INT8 TOPS, while X8 doubles memory capacity and bandwidth to 48 GB and 614.4 GB/s. X8 uses two Siyuan370 chips and provides a dedicated MLU-Link interface. Thus two chips on a board do not justify doubling the published board compute rate. The S4/S8 branch trades peak compute for a 75 W, compact deployment envelope.

Architectural interpretation: the X8 configuration targets data capacity, bandwidth, codec resources and multi-card communication as much as arithmetic density. This can favor larger or bandwidth-sensitive workloads even when peak BF16 matches X4. Fusion helps only when intermediate data and execution dependencies fit the supported mechanism. A portable model graph may still require layout, quantization and kernel tuning to exploit the hardware.

05 Cloud accelerator and board tables

Commercial reference specifications · 1/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Architecture	GA / milestone	Status	Process	Dies / packaging
Siyuan100 / MLU100	n/d	2018	Historical shipping	n/d	n/d
Siyuan270 / MLU270	MLUv02	2019	Historical shipping	n/d	n/d
Siyuan290 / MLU290-M5	n/d	2021 public showcase	Historical shipping	n/d	n/d
MLU370-S4/S8	MLUarch03	2021-2022 product era	Shipping / available	7 nm	SKU configuration
MLU370-X4	MLUarch03	2021-2022 product era	Shipping / available	7 nm	Siyuan370 chip
MLU370-X8	MLUarch03	2021-2022 product era	Shipping / available	7 nm	2 Siyuan370 chips

Commercial reference specifications · 2/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Compute units	Dense BF16	Dense FP8	Dense FP6 / FP4	FP64 vector / matrix
Siyuan100 / MLU100	n/d	n/d	n/d	n/d	n/d
Siyuan270 / MLU270	16 tensor cores	n/d	n/d	n/d	n/d
Siyuan290 / MLU290-M5	n/d	n/d	n/d	n/d	n/d
MLU370-S4/S8	n/d	72 TFLOP/s advertised; dense basis n/d	n/d	n/d	n/d
MLU370-X4	n/d	96 TFLOP/s advertised; dense basis n/d	n/d	n/d	n/d
MLU370-X8	n/d	96 TFLOP/s advertised; dense basis n/d	n/d	n/d	n/d

Commercial reference specifications · 3/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Memory: type; stacks; GB; TB/s	LLC / SRAM
Siyuan100 / MLU100	n/d	n/d
Siyuan270 / MLU270	n/d	n/d
Siyuan290 / MLU290-M5	n/d	n/d
MLU370-S4/S8	LPDDR5; 24/48 GB; 307.2 GB/s	n/d
MLU370-X4	LPDDR5; 24 GB; 307.2 GB/s	n/d
MLU370-X8	LPDDR5; 48 GB; 614.4 GB/s	n/d

Commercial reference specifications · 4/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Scale-up: links; GB/s per direction; domain	Host link	Power / cooling	Form factor
Siyuan100 / MLU100	n/d	n/d	n/d	Cloud accelerator
Siyuan270 / MLU270	n/d	n/d	n/d	Cloud accelerator
Siyuan290 / MLU290-M5	n/d	n/d	n/d	Cloud accelerator
MLU370-S4/S8	n/d	PCIe4 ×16	75 W; passive	Half-height / half-length
MLU370-X4	n/d	PCIe4 ×16	150 W; passive	Full-height full-length; 1 slot
MLU370-X8	4 ports; 100 GB/s one-way aggregate; 8 cards supported	PCIe4 ×16	250 W; passive	Full-height full-length; 2 slots

Host CPU boundary · 1/3

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Generation / reference SKU	Codename	GA	Status	Core µarch	Cores / threads
No proprietary host CPU established	n/a	n/a	n/a	n/a	n/a

Host CPU boundary · 2/3

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Generation / reference SKU	Process per die	Dies / packaging	L2 per core	L3 / domain
No proprietary host CPU established	n/a	n/a	n/a	n/a

Host CPU boundary · 3/3

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Generation / reference SKU	Memory: channels; rate; GB/s	Sockets / links	PCIe / CXL	TDP	Vector / matrix ISA
No proprietary host CPU established	n/a	n/a	n/a	n/a	n/a

06 MLU-Link, systems and the networking roadmap

The X8 board page resolves a key unit issue: 200 GB/s is aggregate bidirectional MLU-Link bandwidth, corresponding to 100 GB/s per direction when divided symmetrically. It lists four ports, sixteen lanes and 50 Gb/s signaling; the board-wide lane/port wording is retained rather than assuming sixteen lanes per port. The page supports an eight-card server configuration but does not establish transparent memory coherence or a fully connected topology.

The 2025 annual report describes future chip platforms for training, language-model inference, multimodal inference and switching for large models. The switching item belongs in the roadmap, not a shipping AI-NIC or DPU table. No port speeds, SerDes generation or switch capacity are assigned without a named technical disclosure. External NICs and host CPUs remain system dependencies in the documented accelerator portfolio.

Networking product boundary · 1/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	GA / milestone	Status	Class
Large-model switch project	2025 annual report	Roadmap	Switch R&D
Merchant DPU / AI NIC	n/a	No product established	n/a

Networking product boundary · 2/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Ports × speed	SerDes	Host interface	Programmable engine
Large-model switch project	n/d	n/d	n/d	n/d
Merchant DPU / AI NIC	n/a	n/a	n/a	n/a

Networking product boundary · 3/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Embedded CPU	On-card memory	Transport / RDMA	Congestion control
Large-model switch project	n/d	n/d	n/d	n/d
Merchant DPU / AI NIC	n/a	n/a	n/a	n/a

Networking product boundary · 4/4

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Product	Key offloads	Power
Large-model switch project	n/d	n/d
Merchant DPU / AI NIC	n/a	n/a

MLU370-X8 links · 1/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Interconnect	Year / generation	Lane rate	Lanes / link	GB/s per link per direction
MLU-Link	MLU370-X8	50 Gb/s	16 total as listed	100 aggregate/device
Host PCIe	MLU370	16 GT/s	16	~31.5 before packet overhead

MLU370-X8 links · 2/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Interconnect	Links / device	Topology	Domain limit	Coherence / semantics
MLU-Link	4 ports	n/d	8-card server supported	Explicit communication; coherence n/d
Host PCIe	1	Point-to-point	Host topology	PCIe

System disclosures · 1/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Platform	Year / milestone	CPU / accelerator count	Scale-up domain
XuanSi1000 / MLU-X1000	WAIC2021	MLU290; count n/d	n/d
MLU370-X8 server	Product documentation	Up to 8 cards	Topology n/d

System disclosures · 2/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Platform	Scale-out per accelerator	Rack power	Cooling
XuanSi1000 / MLU-X1000	n/d	n/d	n/d
MLU370-X8 server	External NIC; n/d	n/d	Passive boards; server airflow

07 Terminal IP, edge products and software continuity

Cambricon1A, 1H and 1M are licensable terminal processor-IP branches, while Siyuan220/MLU220 supplies edge chips and M.2/SOM modules. An IP core inside another company's SoC is not a Cambricon-owned complete SoC. These products share software and instruction-family continuity with cloud offerings, but their memory systems, power budgets and host integration differ. The annual report continues to describe cloud, edge and IP activities without publishing a complete new-generation server SKU matrix.

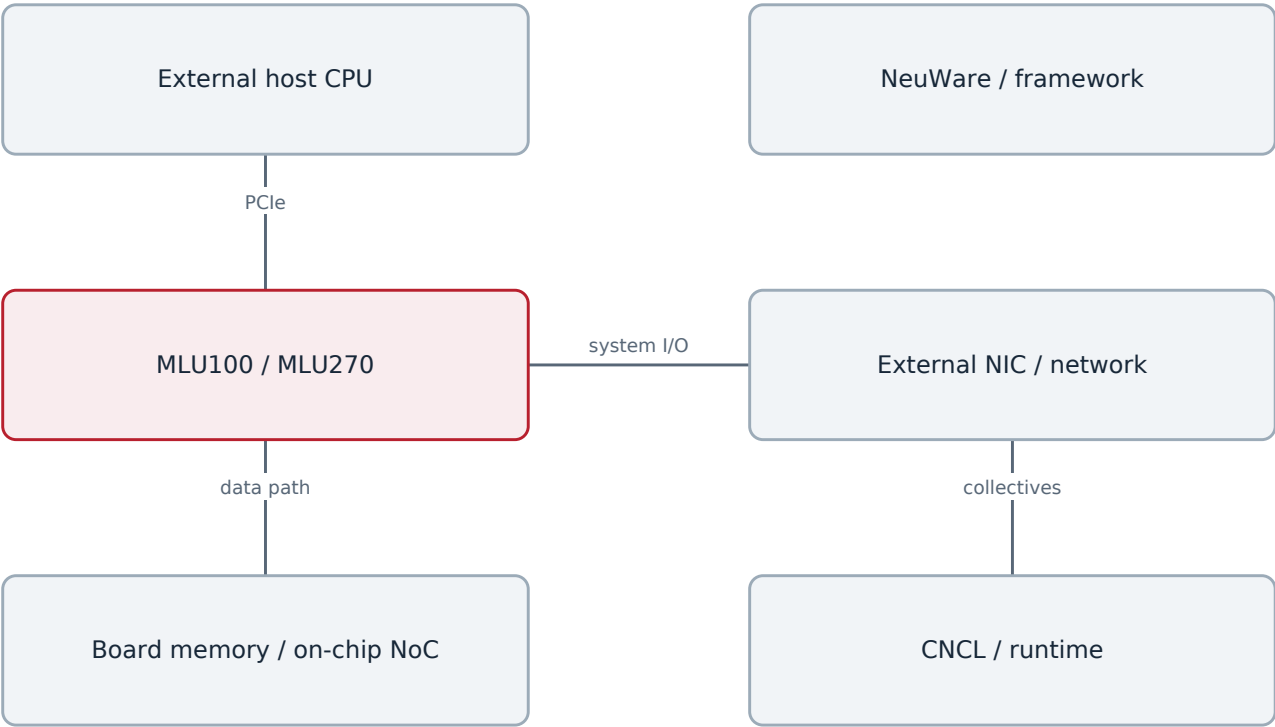
NeuWare brings drivers, runtime, libraries and tools into a shared platform. CNNL implements neural operators, CNCL supplies collective communication and MagicMind provides an MLIR-based inference compilation path. Training supports data, model and hybrid parallelism. Framework support is a starting point: production readiness also requires operator coverage, numerical correctness, communication overlap, profiling and failure recovery at the intended scale.

Software contracts

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

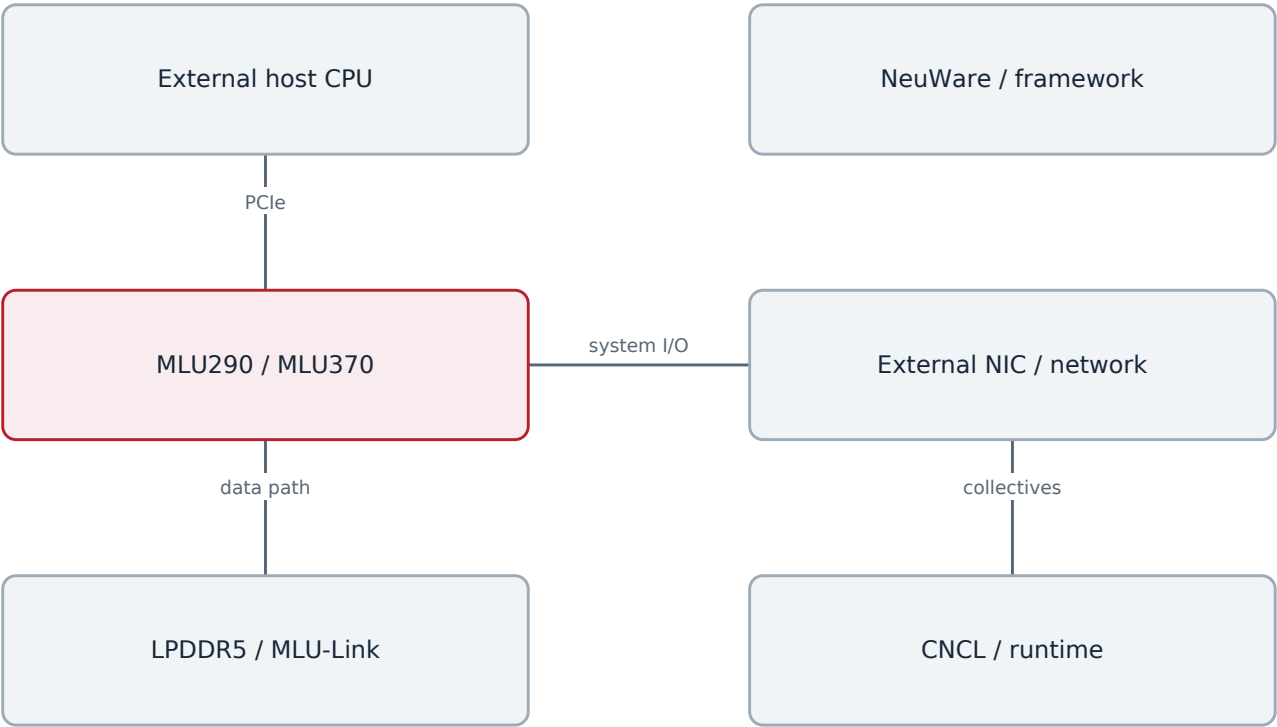
Layer	Product	Function	Scope
Platform	NeuWare	Driver / runtime / tools	Cloud / edge / terminal
Operators	CNNL / BANG	Libraries and kernel programming	Device-specific tuning
Communication	CNCL	Distributed collectives	Topology and configuration dependent
Inference	MagicMind / MLIR	Graph compilation and optimization	Model and operator coverage dependent

2018-2020 stack



Functional stack; MLU290 and MLU370 are separate branches, not one combined configuration.

2021-2026 disclosures stack



Functional stack; MLU290 and MLU370 are separate branches, not one combined configuration.

08 Normalized metrics and deployment implications

Board-level balance

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Metric	Calculation	Basis
X4 DRAM/BF16	$307.2e9/96e12 = 0.0032 \text{ B/FLOP}$	Advertised peak; sparsity basis unstated
X8 DRAM/BF16	$614.4e9/96e12 = 0.0064 \text{ B/FLOP}$	Board-wide; not per chiplet
X8 capacity/BF16	$48e9/96e12 = 5e-4 \text{ B}\cdot\text{s/FLOP}$	LPDDR, not HBM
X8 MLU-Link/DRAM	$100/614.4 = 0.163$	One-way aggregate link rate
X4 vs X8 BF16/TDP	$96/150=0.64$; $96/250=0.384 \text{ TFLOP/s/W}$	Peak/TDP proxy; workload benefit differs
HBM/FLOP; host DDR/core; L3/core	n/d / n/a	No matched published data or own host CPU

The board comparison illustrates why peak performance per watt can mislead: X8 has a lower BF16/TDP ratio than X4 but twice the memory bandwidth and capacity plus direct multi-card communication. A bandwidth-bound or capacity-limited model can therefore favor X8. The correct evaluation includes achievable precision, model fit, host-to-device traffic, inter-card collectives and software versions. Dense floating-point comparisons with other vendors remain conditional where the source does not state the sparsity convention.

09 Conference index and product mapping

Verified disclosure lineage · 1/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Venue	Year	Paper / talk / disclosure
ASPLOS / MICRO	2014	DianNao / DaDianNao
ASPLOS	2015	PuDianNao
ISCA	2015	ShiDianNao
ISCA	2016	Cambricon ISA
MICRO	2016	Cambricon-X
MICRO	2018	Cambricon-S
ISCA	2019	Cambricon-F
ISCA	2021	Cambricon-Q
WAIC	2021-2022	Cloud/edge product exhibitions

Verified disclosure lineage · 2/2

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Venue	Presenter / authors	Product	Evidence type	Source
ASPLOS / MICRO	Paper authors / company	Research baseline	Research record	C01 C03
ASPLOS	Paper authors / company	Research	Research record	C03
ISCA	Paper authors / company	Research	Research record	C03
ISCA	Paper authors / company	Research ISA	Research record	C02
MICRO	Paper authors / company	Research sparsity	Research record	C03
MICRO	Paper authors / company	Research sparsity	Research record	C03
ISCA	Paper authors / company	Research programming architecture	Research record	C04
ISCA	Paper authors / company	Research training	Research record	C03
WAIC	Paper authors / company	MLU220/270/290/370	Product showcase	E03 E04

The big-four architecture record is strongest at ISCA, MICRO and ASPLOS. This edition does not establish a direct commercial MLU implementation paper at HPCA, ISSCC or Hot Chips. It also does not treat a same-name academic paper as proof of a commercial design. Recent annual reports describe large-model development and deployment but omit a full named implementation matrix. Accordingly, widely circulated newer model numbers are not assigned speculative process, memory or throughput specifications here; this does not imply that MLU370 is the company’s newest deployed silicon.

Naming boundaries

n/d: not established; n/a: not applicable. Precision, scope and direction are shown in the cells; sources are mapped at the end.

Name	Meaning	Boundary
DianNao family	Academic designs	Not commercial MLU SKUs
Siyuan / MLU	Chip and board families	Always specify board suffix
MLU370 chiplet / X8 dual-chip	Package / board integration	Different counting levels
MLU-Link / CNCL	Hardware links / collective library	Bandwidth versus software behavior

Source map

- 01 Scope and architectural synthesis - C02, C03, E02, E03, P01, P02
- 02 Research lineage: what the architecture conferences establish - C01, C02, C03, C04, C05
- 03 Commercial generations: MLU100, 270 and 290 - E02, E03, E05, P01
- 04 MLU370: chiplets, fusion and board differentiation - P02, P04, P05, P06, P07
- 05 Cloud accelerator and board tables - E02, E03, E04, E05, P01, P02, P04, P05, P06
- 06 MLU-Link, systems and the networking roadmap - E02, E03, P05, P06
- 07 Terminal IP, edge products and software continuity - E02, E03, E05, P01, P02, P06, P07
- 08 Normalized metrics and deployment implications - E02, P05, P06
- 09 Conference index and product mapping - C01, C02, C03, C04, E02, E03, E04, P02, P06, P07

References

Product and technical documentation

[P01] Siyuan270

<https://cambricon.com/index.php?a=lists&c=index&catid=15&m=content>

Primary source; accessed by 2026-09-28

[P02] Siyuan370

<https://cambricon.com/index.php?a=lists&c=index&catid=360&m=content>

Primary source; accessed by 2026-09-28

[P04] MLU370-S4/S8 board

<https://www.cambricon.com/index.php?m=content&c=index&a=lists&catid=365>

Primary source; accessed by 2026-09-28

[P05] MLU370-X4 board

<https://www.cambricon.com/index.php?m=content&c=index&a=lists&catid=371>

Primary source; accessed by 2026-09-28

[P06] MLU370-X8 board

<https://www.cambricon.com/index.php?m=content&c=index&a=lists&catid=406>

Primary source; accessed by 2026-09-28

[P07] Cambricon NeuWare

<https://www.cambricon.com/index.php?m=content&c=index&a=lists&catid=71>

Primary source; accessed by 2026-09-28

Conference records and implementation disclosures

[C01] DianNao project and publication lineage

<https://novel.ict.ac.cn/diannao/>

Primary source; accessed by 2026-09-28

[C02] Cambricon ISA, ISCA2016 author record

<https://researchportal.hkust.edu.hk/en/publications/cambricon-an-instruction-set-architecture-for-neural-networks/>

Primary source; accessed by 2026-09-28

[C03] Tianshi Chen publications

<https://novel.ict.ac.cn/tchen/>

Primary source; accessed by 2026-09-28

[C04] Cambricon-F, ISCA2019 full author paper

<https://dl.yongwei.site/F.pdf>

Primary source; accessed by 2026-09-28

[C05] Cambricon-LLM research, 2024

<https://arxiv.org/abs/2409.15654>

Primary source; accessed by 2026-09-28

Launches, ecosystem and deployment

[E02] Cambricon2025 annual report, company filing mirror

https://money.finance.sina.com.cn/corp/view/vCB_AllBulletinDetail.php?id=11993143&stockid=688256

Primary source; accessed by 2026-09-28

[E03] WAIC2021 portfolio

<https://www.cambricon.com/index.php?a=show&c=index&catid=127&id=41&m=content>

Primary source; accessed by 2026-09-28

[E04] WAIC2022 portfolio

<https://www.cambricon.com/index.php?a=show&c=index&catid=127&id=51&m=content>

Primary source; accessed by 2026-09-28

[E05] Siyuan220 launch

<https://cambricon.com/index.php?a=show&c=index&catid=127&id=11&m=content>

Primary source; accessed by 2026-09-28